

100 BASIC TECHNOLOGY QUESTIONS & ANSWERS

Hardware & Devices

1. What does CPU stand for?

Central Processing Unit, often called the "brain" of the computer.

2. What is RAM?

Random Access Memory; it provides short-term working space for active programs.

3. What is the difference between an HDD and an SSD?

An HDD uses spinning magnetic disks for storage, while an SSD uses faster, non-moving flash memory chips.

4. What is a motherboard?

The main circuit board that connects all physical components of a computer.

5. What does GPU do?

The Graphics Processing Unit handles rendering images, video, and 2D/3D graphics.

6. What is a peripheral device?

An external device connected to a computer, like a mouse, keyboard, or printer.

7. What does USB stand for?

Universal Serial Bus, a standard connection for power and data transfer.

8. What is the function of a power supply unit (PSU)?

It converts electrical power from an outlet into usable power for the computer's components.

9. What is Bluetooth?

A short-range wireless technology used to connect devices like headphones and mice.

10. What does HDMI stand for?

High-Definition Multimedia Interface, used for transmitting audio and video to monitors and TVs.

11. What is a pixel?

The smallest single point of color on a digital display.

12. What is a server?

A powerful computer dedicated to managing network resources and hosting data or websites.

Software & Operating Systems

13. What is an Operating System (OS)?

The primary software that manages hardware and allows other programs to run (e.g., Windows, macOS).

14. What is a software driver?

A program that allows an operating system to communicate with a specific hardware device.

15. What is open-source software?

Software whose source code is publicly available for anyone to view, modify, and distribute.

16. What is a software bug?

An error or flaw in a computer program that causes it to behave unexpectedly.

17. What is an application (app)?

A specific software program designed to perform a particular task for the user.

18. What does GUI mean?

Graphical User Interface, which allows users to interact with devices through icons and visual indicators.

19. What is cloud computing?

Storing and accessing data and programs over the internet instead of on a computer's hard drive.

20. What is a software patch?

An update designed to fix bugs, close security vulnerabilities, or improve software.

21. What does SaaS stand for?

Software as a Service, a model where software is licensed on a subscription basis and hosted centrally.

22. What is an algorithm?

A set of step-by-step instructions or rules for a computer to solve a problem.

23. What is source code?

The human-readable instructions written by a programmer that make up a software application.

24. What does compiling mean?

Translating source code into machine-readable code that a computer can execute.

25. What is a beta version?

A pre-release version of software given to a group of users to test for bugs.

26. What is cache memory?

A small, fast memory area used to temporarily store frequently accessed data for quick retrieval.

Internet & Networking

27. What is an IP address?

Internet Protocol address; a unique string of numbers identifying a device on a network.

28. What does URL stand for?

Uniform Resource Locator, which is the web address you type into a browser.

29. What is a router?

A device that directs internet traffic and connects multiple devices to a single network.

30. What is a modem?

A device that connects your home network to your Internet Service Provider.

31. What is an ISP?

Internet Service Provider, a company that provides access to the internet.

32. What is Wi-Fi?

A wireless networking technology that allows devices to communicate over radio frequencies.

33. What is bandwidth?

The maximum rate of data transfer across a network path.

34. What is latency or "ping"?

The time it takes for data to travel from one point on a network to another.

35. What is the difference between HTTP and HTTPS?

HTTPS includes a security protocol (encryption) to protect data transmitted between a browser and a website.

36. What is DNS?

Domain Name System; it translates human-readable web addresses into machine-readable IP addresses.

37. What is a web browser?

A software application used to access and view websites (e.g., Chrome, Safari).

38. What is a search engine?

A website used to search the internet for information (e.g., Google, Bing).

39. What is an Ethernet cable?

A physical cable used to connect devices on a local wired network for stable internet.

40. What does HTML stand for?

HyperText Markup Language, the standard code used to structure content on web pages.

41. What is a domain name?

The easy-to-remember name of a website, such as "example.com".

42. What is a web server?

A computer system that hosts websites and serves web pages to users upon request.

43. What is the "cloud"?

A global network of remote servers operating together to store and manage data.

44. What is a cookie in web browsing?

A small text file saved by a website to remember your preferences or login status.

45. What is a firewall?

A network security system that monitors and controls incoming and outgoing network traffic.

46. What does 5G mean?

The 5th generation of cellular network technology, offering faster mobile internet speeds.

Cybersecurity & Privacy

47. What is malware?

Malicious software designed to harm, exploit, or disable computers and networks.

48. What is a computer virus?

A type of malware that attaches itself to legitimate code and spreads from device to device.

49. What is a Trojan horse?

Malware disguised as legitimate software to trick users into installing it.

50. What is phishing?

A fraudulent attempt to obtain sensitive information by disguising as a trustworthy entity in electronic communications.

51. What is ransomware?

Malware that encrypts a victim's files, with the attacker demanding payment to unlock them.

52. What is encryption?

The process of converting readable data into an unreadable format to prevent unauthorized access.

53. What is a VPN?

Virtual Private Network; it creates a secure, encrypted connection over a less secure network like the internet.

54. What is Two-Factor Authentication (2FA)?

A security process requiring two different forms of identification before granting access to an account.

55. What is a password manager?

An application that securely stores and manages all your passwords in an encrypted database.

56. What is a data breach?

An incident where sensitive, protected, or confidential data is copied, transmitted, or stolen.

57. What is spyware?

Software installed on a device without the user's knowledge to track their activities.

58. What is DDoS?

Distributed Denial of Service; an attack meant to overwhelm a website with fake traffic so it crashes.

59. What does it mean to clear your cache?

Deleting temporary internet files to resolve browser errors or free up space.

60. What is incognito or private browsing mode?

A browser setting that prevents the storing of history, cookies, and site data on the local device.

61. What is biometrics?

Using physical characteristics (like fingerprints or facial recognition) for security authentication.

62. What is adware?

Software that automatically displays or downloads advertising material when a user is online.

63. What is a software backup?

A copy of computer data taken and stored elsewhere so it can be restored if the original is lost.

64. What is end-to-end encryption?

A system where only the communicating users can read the messages, keeping data safe from third parties.

File Types & Formats

65. What is a PDF?

Portable Document Format; a file type used to present documents dependably across any device or OS.

66. What is a JPEG (or JPG)?

A commonly used method of lossy compression for digital images, resulting in smaller file sizes.

67. What is an MP4?

A digital multimedia container format most commonly used to store video and audio.

68. What is an MP3?

A standard audio coding format for digital audio that compresses sound data to save space.

69. What is a ZIP file?

An archive file format that supports lossless data compression, allowing multiple files to be bundled together.

70. What is a CSV file?

Comma-Separated Values; a plain text file that contains a list of data, often used in spreadsheets.

71. What does GIF stand for?

Graphics Interchange Format, commonly used for simple, short animated images on the web.

Emerging Tech & General Concepts

72. What is Artificial Intelligence (AI)?

The simulation of human intelligence processes by machines, especially computer systems.

73. What is Machine Learning?

A subset of AI that allows software applications to become more accurate at predicting outcomes without explicit programming.

74. What is the Internet of Things (IoT)?

The network of physical objects embedded with sensors and software that connect and exchange data over the internet.

75. What is Blockchain?

A decentralized, distributed digital ledger used to record transactions securely across many computers.

76. What is Cryptocurrency?

A digital or virtual currency that is secured by cryptography and operates independently of a central bank.

77. What is Virtual Reality (VR)?

A simulated 3D environment that users can explore and interact with via specialized headsets.

78. What is Augmented Reality (AR)?

An interactive experience where real-world environments are enhanced by computer-generated visual overlays.

79. What is an NFT?

Non-Fungible Token; a unique digital identifier recorded on a blockchain used to certify ownership and authenticity.

80. What is Big Data?

Extremely large, complex data sets that require specialized computational systems to analyze for trends and patterns.

81. What does UI stand for?

User Interface; the visual elements (buttons, menus) a user interacts with on a screen.

82. What does UX stand for?

User Experience; the overall experience and satisfaction a user has when interacting with a product or system.

83. What is a smart home device?

An electronic device connected to the internet that can be controlled remotely or automated (e.g., smart thermostat).

84. What is a chatbot?

A software program designed to simulate conversation with human users, often using AI.

85. What is data mining?

The practice of analyzing large databases in order to generate new information or discover patterns.

Troubleshooting & Operations

86. What does it mean to reboot a computer?

To completely shut down the operating system and hardware, and then start it up again.

87. What is safe mode?

A diagnostic startup mode in operating systems used to troubleshoot system issues by loading only basic drivers.

88. Why is my computer running slowly?

Often due to full hard drives, too many active applications, lack of RAM, or malware infections.

89. What is task manager?

A system monitor program used to provide information about the processes and programs running on a computer.

90. What does "Ctrl+Alt+Del" do on Windows?

It interrupts the OS and opens a menu allowing users to lock the PC, switch users, or open Task Manager.

91. How do you force quit an application on Mac?

By pressing "Command + Option + Escape" and selecting the unresponsive application.

92. What is a blue screen of death (BSOD)?

An error screen displayed on Windows computers following a fatal system error.

93. What does "plug and play" mean?

Hardware that automatically begins working as soon as it is plugged into a computer, without needing manual configuration.

94. What should you do if an application freezes?

Wait a moment, attempt to close it normally, and if it fails, use the system's task manager to force quit the program.

95. What is the recycle bin or trash?

A temporary holding area for files deleted by the user, which can still be restored until the bin is emptied.

96. What is a hard reset?

Restoring a device to its original factory settings, erasing all user data in the process.

97. What is clipboard in computing?

A temporary storage area for data that the user wants to copy from one place to another.

98. How do you take a screenshot on most Windows PCs?

By pressing the "Print Screen" (PrtScn) key, or using the Snipping Tool.

99. How do you take a screenshot on a Mac?

By pressing "Command + Shift + 3" (for the whole screen) or "Command + Shift + 4" (to select an area).

100. Why is software updating important?

Updates patch security holes, fix bugs, and provide new features, keeping the system safe and functional.